

The fts-workspace guides, guide 6 of 7

Guide 6: Fluid-structure interaction

Fluid-structure interaction, as pyfs does it

Written for pyflightstream 0.30.0.

Sources: the package page “FSI in a workspace”, the FSI tutorial and the FSI examples [1].

pyflightstream is the work of Geovana Neves: <https://github.com/nevesgeovana/pyflightstream>, concept DOI 10.5281/zenodo.21482924.

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What FSI means here, and how to read this deck

- **FSI** couples the aerodynamic loads on a blade or a wing to its structural deflection: FlightStream solves the flow, pyflightstream solves the structure, and the solver deforms its surface with the displacements it is handed.
- The structure is **one Euler beam per blade**, or one along the wing (PyNite [5]), in flap and torsion, solved quasi-statically at every coupling call.
- A matrix row selects its structure by name, FSI: f001, and calibrates it from the matrix.
- Each frame names its source at its foot. The example numbers are what the package's own examples print, on synthetic blades, and one campaign on a public propeller geometry [2].

The optional extra

The beam needs PyNite, from the [fsi] extra: `pip install pyflightstream[fsi]`. Without it the package says how to install it.

***Takeaway** FSI runs on steady, `qsteady_rotor` (sector) and `unsteady` rows. A turning rotor, `unsteady_rotor`, is refused: that coupling is still being debugged.*

Source: pyflightstream [1]; FSI tutorial (src/pyflightstream/fsi/README.md); fsi/beam.py; extras.py; the page "FSI in a workspace".

Where FSI runs, and what the structure carries

Workflow	FSI	What the structure carries, and why
steady	allowed	a fixed wing: the aerodynamic loads and the structure's own weight ; nothing turns, so no centrifugal load
qsteady_rotor, sector	allowed	one blade held still in a free stream turning at the rotor's speed: the structural solver applies the centrifugal loads and stiffening at that speed, Ω taken from the row's free-stream rotation
qsteady_rotor, wheel	refused	the rigid wheel runs; its coupled form is not offered
unsteady	allowed	a structure that does not turn: the loads and its own weight, one coupling call per time step
unsteady_rotor	refused	the plan says the feature is still in debug: on 26.124 the solver deforms a turning blade in its un-turned position [4]

A quasi-steady answer is valid only for an isolated, axisymmetric rotor, and only while the flow at each station changes slowly beside the revolution (Guide 5, the reduced frequency).

Source: the requirements of this version; `cases/fsi_workspace.py`; `cases/workflows.py`.

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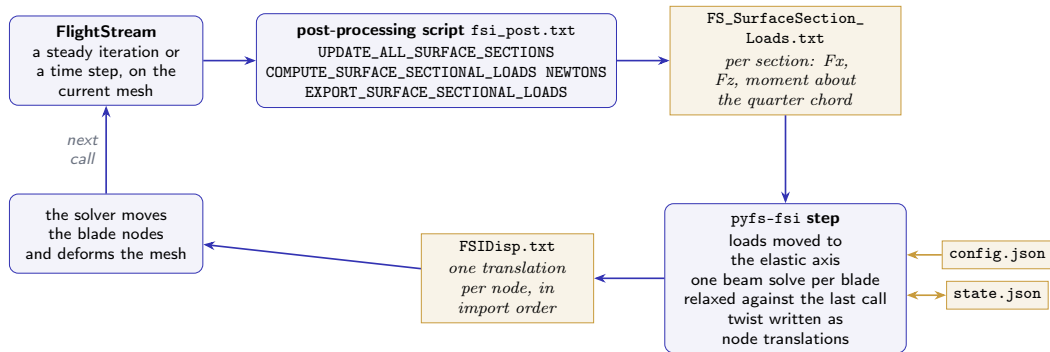
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Part 1

The coupling loop

The loop, one coupling call

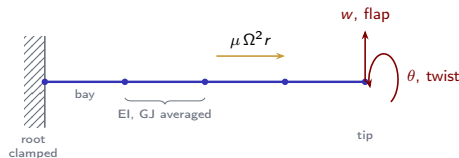


FlightStream's Aeroelastic Toolbox calls the executable once per aeroelastic iteration of a steady row, or once per time step of an unsteady one (`SET_AEROELASTIC_ITERATIONS 1`). Each call is stateless: what persists is in `state.json`. Loads and displacements are in each blade's own frame, or the wing's; the package never sees azimuth.

Source: `pyflightstream` [1]; FSI tutorial, "The loop in one paragraph", "cli"; `cases/fsi_workspace.py`, `wire_workspace_fsi`; `fsi/__init__.py`.

The structure: one Euler beam per blade, or along the wing

- One node per radial station, on the elastic axis; one member per bay, with the bay-averaged EI and GJ; the root station clamped.
- Unit elastic moduli, $E = G = 1$: the section constants handed to PyNite are the configuration's EI and GJ, so no modulus is applied twice.
- Flap deflection w and elastic twist θ carry mass (μI and $(I_1 + I_2) I$); every other degree of freedom is condensed out of the eigenproblem exactly.
- Static solves are linear, or P-Delta whenever axial tension is present [6, 7].



Takeaway The stiffnesses in *config.json* are the beam's stiffnesses, typed or generated, exactly as written.

Source: pyflightstream [1]; FSI tutorial, "beam"; fsi/beam.py module docstring.

What rotation adds

- **Tension.** The distributed axial force $\mu \Omega^2 r$ makes the tension $N(r)$ emerge in the solver, and P-Delta turns it into bending stiffness, with no manual correction terms: the centrifugal stiffening of a rotating blade [8, 9, 10].
- **Propeller moment.** $-\Omega^2(l_1 - l_2) \sin \theta \cos \theta$ drives each section toward flat pitch. It depends on the total pitch, geometric plus elastic, so each call re-solves the beam a few times until the twist stops moving [9, 12].
- **In-plane softening.** The part of the flap deflection that lies in the plane of rotation feels the centrifugal field pulling it outward, a term $\mu \Omega^2 \sin^2 \beta w$, β the section's pitch, that lowers the stiffness; it is applied with the tension [8, 9].
- **Not modelled: the trapeze effect.** A declared limitation; omitting it overpredicts the elastic twist, which is conservative.

Source: pyflightstream [1]; FSI tutorial, “The loop in one paragraph” and “centrifugal”; `fsi/centrifugal.py` module docstring.

A fixed wing: its own weight, and where the model stops

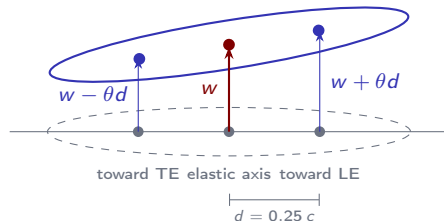
- **A wing that does not turn** has no centrifugal load. The structural solver applies the structure's **own weight**, gravity on its mass distribution, beside the aerodynamic loads, on a steady or an unsteady row; the loop is the same.
- **A blade held still** in a turning free stream (`qsteady_rotor`, `sector`) is the opposite case: the aerodynamics see no motion, and the structural solver still applies the centrifugal loads at the row's rotation speed.
- **Quasi-static.** The mean deformation is exact within the beam idealisation; the once-per-revolution content is trustworthy only well below resonance, roughly $n\Omega/\omega_n \leq 0.3$ [12, 13]. The convergence log prints this boundary in its header.

The formulas are transcribed, not re-verified

The primary sources of the model have not been independently checked against the implemented formulas; each formula lives in one small function that cites its source, so a correction stays local.

Source: the requirements of this version; `fsi/centrifugal.py`; the page "FSI in a workspace".

Twist travels as three nodes per station



- FSIDisp.txt carries translations only, so twist is encoded by three nodes per station, $d = 0.25 c$ apart (node_offset_chord_fraction).
- Rigid section kinematics, $\delta y = w + \theta d$: linear, so the inverse is exact to machine precision.
- One generator makes the node file, the ordering map and the FSIDisp.txt rows: the solver applies them strictly in import order, and a drift would corrupt the geometry in silence.
- One node file serves every blade, imported once per blade frame, in blade order.

Source: pyflightstream [1]; FSI tutorial, "kinematics" and "nodes"; fsi/nodes.py module docstring; fsi/config.py, FsiConfig.node_offset_chord_fraction.

The driver: four phases, keyed on the step counter

Phase	Behaviour, and its key in <code>config.json</code> phases
1, wake development	zero displacements on the rigid blade; <code>wake_development_revolutions</code> (1.0)
2, averaged coupling	loads averaged over the window, relaxed updates $d_{new} = d_{old} + \lambda(d_{calc} - d_{old})$; <code>averaging_window_revolutions</code> (1.0), <code>coupling_relaxation</code> (0.4)
3, convergence watch	as 2, and converged when BOTH hold across consecutive revolutions: tip twist change below <code>tip_twist_tolerance_deg</code> (0.05) and relative change of the integrated normal force below <code>thrust_tolerance_fraction</code> (0.02)
4, recording	instantaneous loads, $\lambda = 1$ by design, twist recorded per step; <code>recording_revolutions</code> (1.0)

- **Freshness is asserted:** each call must see an advancing solver iteration in the loads header; a repeat is refused, since it means more than one FSI iteration per time step.
- **Frozen mode:** a `fsi_frozen_displacements.txt` in the run folder is replayed verbatim at every call, with no loads and no solve.

Source: `pyflightstream` [1]; FSI tutorial, “driver and state”; `fsi/config.py`, `PhaseSchedule` (defaults in brackets).

The structural nodes, and how the surface follows them

- **Inside the component.** The nodes lie on the camber line of each station, turned by its local twist, inside the blade or the wing; a node outside is refused at plan.
- **In the frame that turns with the blade,** so a displacement means the same thing at every position.
- **The surface by its boundary:** the aeroelastic surface is assigned by the boundary's own identifier in the mesh.
- **After the structural call:** a coupled point exports once the last call has moved the surface; a steady run waits for it.

A radial basis function carries the nodes to the panels

The displacement known at the nodes is interpolated onto every aerodynamic panel [14, 15, 16]. A beam is a **line** of nodes: a compactly supported kernel reaches only the panels within its radius of that line, and the rest of the surface barely moves. The coupled row uses the **multiquadric** [14], whose support is global.

Source: the requirements of this version; `fsi/nodes.py`; `cases/fsi_workspace.py`; Guide 4, `aeroelastic_rbf_type`.

Route B: the direct transform, a sanity check

- The same beam, solved on the solver's sectional loads, gives flap $w(r)$ and twist $\theta(r)$; **every mesh vertex is moved** about the elastic axis by that w and θ , and the deformed geometry is solved again, rigid. No interpolation, no in-run coupling.
- Repeated until the tip flap and twist stop moving, it is a partitioned static iteration: quasi-steady by construction.
- A zero deformation through the same chain reproduces the rigid answer, which is its own control.

What route B is for

A simplified run, a check of the coupled route against an independent one, and an FSI option where the coupled route is not available. It is **not** the coupled route: it carries no dynamics and no in-run deformation. The worked example of Part 7 was solved this way.

Source: the requirements of this version; the worked example's route diagnostic [3].

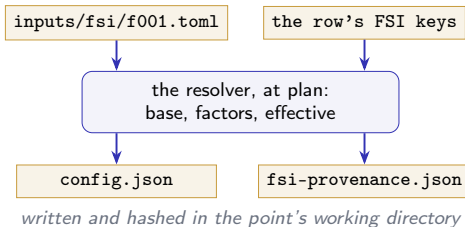
Part 2

The structural input

One structural input, selected by name

- One configuration per file, `inputs/fsi/f001.toml`: `f`, then letters, digits, `_` or `-`; no extension in the row.
- The row selects it in `VAR_NAMES_VALUES`, with any calibration beside it.
- The resolver attaches the *effective* `FsiConfig` and its provenance to the case. No solver runs while loading or planning.
- The run writes both files in the point's working directory and records their hashes. The source is hashed too: changing it means planning again.

FSI: f001 / FSI_BENDING_STIFFNESS_N_M2_FACTOR: 1.15



Takeaway *The geometry library is never a destination for these generated files.*

Source: `pyflightstream` [1]; `docs/fsi-workspace.md`, opening section; `workspace/fsi_setup.py`, `resolve_row_fsi`; `docs/srs/data-model.md`, the `inputs/` tree.

Two ways to state the blade

	mode = "supplied"	mode = "calculated"
What you write	the FsiConfig scalars under [config] and the complete BladeProperties arrays under [config.blade]	the [config] scalars, a material, and [sections]: one closed contour per station, in metres
What the package computes	nothing: the distributions are the file's, then calibrated	area, centroid, principal inertias, the torsion constant, then EI, GJ, running mass and mass inertias from the material
Where the numbers come from	you: the example distributions are synthetic, not a material or airframe dataset	the geometry and ONE sourced material, recorded with the numbers
Refused	a material or [sections] table; any material factor	a [config.blade] table; no material; no [sections]

Takeaway *Either way the result is the existing solid homogeneous Euler beam: this input adds no shell, spar, hollow-section, shear-centre or second structural model.*

Source: pyflightstream [1]; docs/fsi-workspace.md, "Supplied distributions" and "Calculated solid sections"; workspace/fsi_setup.py, resolve_fsi_setup.

Supplied distributions

```
mode = "supplied"

[calibration]
bending_stiffness_n_m2 = 1.05

[config]
blade_count = 2
omega_rad_per_s = 0.0

[config.blade]
station_radii_m = [0.1, 0.5]
chord_m = [0.04, 0.04]
mass_per_length_kg_per_m = [1.0, 2.0]
inertia_major_kg_m = [0.001, 0.002]
inertia_minor_kg_m = [0.0001, 0.0002]
bending_stiffness_n_m2 = [10.0, 20.0]
torsion_stiffness_n_m2 = [5.0, 10.0]
elastic_axis_offset_chordwise_m = [0.0, 0.0]
elastic_axis_offset_normal_m = [0.0, 0.0]
cg_offset_chordwise_m = [0.0, 0.0]
cg_offset_normal_m = [0.0, 0.0]
geometric_pitch_deg = [0.0, 0.0]
```

Array	Unit
station_radii_m, strictly increasing	m
mass_per_length_kg_per_m	kg/m
bending_stiffness_n_m2 (EI, flap)	N m ²
torsion_stiffness_n_m2 (GJ)	N m ²
inertia_major_kg_m, _minor_kg_m	kg m
offsets, chord_m	m

- Every array has the length of `station_radii_m`; values are finite; mass, EI, GJ and chord positive; principal inertias not negative.
- With the row factor 1.15 of the previous part, the effective EI is [11.5, 23.0]: the row replaces the file's 1.05, it does not multiply it.

A zero-speed input like this one is a valid structural input for offline checks, not a runnable coupled rotor row.

Source: `pyflightstream [1]; docs/fsi-workspace.md, "Supplied distributions"; fsi/config.py, BladeProperties.`

Calculated solid sections

```
mode = "calculated"
material = "ti-6al-4v-grade5-annealed"

[config]
blade_count = 2
omega_rad_per_s = 0.0

[sections]
station_radii_m = [0.1, 0.5]
chord_m = [0.04, 0.04]
geometric_pitch_deg = [0.0, 0.0]
geometry_source = "Synthetic rectangular sections in SI
units"
torsion_grid_cells = 16
sections_m = [
    [[-0.02, -0.002], [0.02, -0.002],
     [0.02, 0.002], [-0.02, 0.002]],
    [[-0.02, -0.002], [0.02, -0.002],
     [0.02, 0.002], [-0.02, 0.002]]
]
```

- `sections_m`: one closed polygon per station, chordwise and normal metres about the pitch axis.
- `geometry_source`: what the contours were built from, in words; it travels with the numbers.
- `torsion_grid_cells`: cells across the thickness for the torsion solve, 8 to 512, 64 when omitted.
- No `[config.blade]`: this mode derives it.

Takeaway A 40 by 4 mm titanium strip; example 1 turns these contours into EI.

Source: `pyflightstream` [1]; `docs/fsi-workspace.md`, "Calculated solid sections"; `workspace/fsi_setup.py`, `SolidSections`, `FSI_TEMPLATE`.

The material database: no number without a source

Key	ρ [kg/m ³]	E [GPa]	G [GPa]	ν	Source, as recorded
ti-6al-4v-grade5-annealed	4430	113.8	44.0	0.342	ASM Aerospace Specification Metals data sheet, annealed; G tabulated
al-7075-t6	2810	71.7	26.9	0.33	ASM Aerospace Specification Metals data sheet, T6 typical values; G tabulated

- Each entry's four numbers come from ONE cited data set, with its table, condition, address and reading date. Database version 1; a changed number is a new version.
- Your own material: a [material] table stating every field, with a [material.source] of document, table, condition, url, consulted. Density, E , G finite and positive; $-1 < \nu < 0.5$.
- Room-temperature typical values for a structural model of a blade, not the design allowables of a certified part.

Source: `pyflightstream` [1]; `fsi/materials.py`; `docs/fsi-workspace.md`, "Calculated solid sections"; FSI tutorial, "sections and materials".

The solid-section calculator

Field	Generated as
mass_per_length_kg_per_m	ρA
inertia_major_kg_m, _minor	ρ times the larger, smaller principal second moment
bending_stiffness_n_m2	E times the second moment about the chordwise centroidal axis
torsion_stiffness_n_m2	GJ , J computed numerically
elastic_axis_offset_*	the centroid's position from the pitch axis
cg_offset_*	zero

- Area, centroid, second moments: exact closed forms of the polygon. A contour that crosses or touches itself, repeats a vertex or folds back is refused by name.
- J : the Prandtl stress function by finite differences cut exactly at the polygon [6, 7]. 16 cells across are within about 1 % of the closed forms, 32 within 0.3 %, 64 within 0.08 %.
- The thin-section $\frac{1}{3} \int t^3 ds$ is kept as a cross-check, never as the value.

Stated hypotheses, recorded with every generated blade

SOLID, HOMOGENEOUS sections; the elastic axis at the CENTROID, the shear centre not computed; a zero elastic-axis-to-CG offset, so no bend-twist coupling from it; flap bending about the chordwise axis.

Source: pyflightstream [1]; FSI tutorial, "sections and materials".

Part 3

Calibration and provenance

Calibration factors: unity, the file, the matrix

Level	Property names
Distributions, either mode	mass_per_length_kg_per_m, inertia_major_kg_m, inertia_minor_kg_m, bending_stiffness_n_m2, torsion_stiffness_n_m2
Orientation	geometric_pitch_deg
Offsets	elastic_axis_offset_chordwise_m, _normal_m, cg_offset_chordwise_m, _normal_m
Material, calculated mode only	density_kg_per_m3, youngs_modulus_pa, shear_modulus_pa, poisson_ratio

The matrix key is FSI_ + the name in capitals + _FACTOR: FSI_TORSION_STIFFNESS_N_M2_FACTOR.

matrix: FSI_*_FACTOR
on the row, even 1.0

file: the TOML's
[calibration] table

unity: stated nowhere

takes precedence ↑

- Every factor is dimensionless, finite and positive.
- An explicit matrix value replaces the file factor exactly once; it is never multiplied by it.

Source: pyflightstream [1]; docs/fsi-workspace.md, "Calibration and provenance"; _fsi_calibration.py, MATRIX_FACTORS.

One property, scaled once

Calibrating a source and a property derived from it would scale the same number twice. Refused, in calculated mode:

Source factor	with any of
density	mass per length, either mass inertia
Young's modulus E	EI
shear modulus G	GJ
E or ν , when G is derived	a separate G or GJ factor

“...: calibrating `youngs_modulus_pa` and its derived `bending_stiffness_n_m2` would scale the same property twice; choose one calibration level.”

- The legacy `config.stiffness_scale_factor` must be 1.0 in a workspace input: calibrate EI and GJ explicitly instead of a second scaling at solve time.
- Supplied mode takes no material factor: calibrate the distribution itself.
- Every effective property is validated again, including major principal inertia not below minor.

File 1.05, row 1.15

Effective EI = base $\times$ 1.15, not base $\times$ 1.2075. The origin recorded is `matrix`.

Source: `pyflightstream` [1]; `workspace/fsi_setup.py`, `resolve_fsi_setup`; `docs/fsi-workspace.md`, “Calibration and provenance”.

A tabulated shear modulus stays tabulated

- A material either takes G as its source tabulates it, or derives it as $E/(2(1+\nu))$ and says so (`shear_modulus_basis`).
- When the source tabulates G , changing E or ν keeps that G . Only a material declaring a derived G follows the formula.
- The two contracts stay distinct even at unity calibration: nothing substitutes the isotropic formula for a tabulated value.
- Calibration never rewrites the source database; the base and effective material both go to the provenance.

The titanium entry

Tabulated: $G = 44.0$ GPa.

From the same sheet, $E/(2(1+\nu)) = 42.4$ GPa.

The entry keeps 44.0, about 4% apart, and a torsion stiffness built from it keeps it too.

Takeaway *Unity calibration, or a density, E or ν change, does not substitute the formula for a tabulated G .*

Source: `pyflightstream` [1]; `docs/fsi-workspace.md`, “Calculated solid sections” and “Calibration and provenance”; `fsi/materials.py`.

The pitch factor scales angles, once

- `FSI_GEOMETRIC_PITCH_DEG_FACTOR` scales each signed structural angle in degrees once; zero stays zero.
- It changes the orientation of the spinning beam's sections used to generate the structural nodes and to project the loads; section properties, radii and node order stay.
- It is not an angular offset. It does not turn the section contours again, and it does not re-pitch the aerodynamic mesh.
- Keep the aerodynamic geometry consistent with the intended structural orientation, and regenerate the node file and ordering map from the effective configuration.

File factor 2.0, row factor 1.5

```
geometric_pitch_deg = [-60,  
20]  
  
becomes [-90, 30],  
not [-180, 60].
```

Alone, it validates nothing

This parameter alone does not establish coupled aerodynamic validation.

Source: `pyflightstream [1]; docs/fsi-workspace.md`, "Calibration and provenance".

Provenance: every number says where it came from

`fsi-provenance.json`, per point

- the source path and its sha256, and the mode;
- the unscaled (base) and effective configurations;
- every factor, and whether it came from unity, file or matrix;
- calculated mode: the base and effective material properties and the section algorithm's provenance;
- `section_model`: "existing solid homogeneous Euler beam".

Inside a generated blade:

`BladeProperties.provenance`

- the material entry, its source and the database version;
- the geometry file's NAME and sha256, never its path;
- the hypotheses, the torsion method and its grid per station, and the thin-section ratio per station.

`config_sha256`

- over the sorted, whitespace-free JSON: it identifies the physics, not the file layout;
- on every row of the convergence log. A typed configuration has no provenance and keeps the digest it always had.

Source: `pyflightstream` [1]; `workspace/fsi_setup.py`, `ResolvedFsiSetup.provenance`; `fsi/config.py`, `config_sha256`; FSI tutorial, "config".

Part 4

From the row to the solver

What a coupled row stages

A row that states FSI: f001 on a workflow that allows it prepares the driver's complete input interface. Planning derives the nodes and the maps from the effective configuration and resolves the boundaries and the frames of each blade or of the wing; the run writes and hashes:

File, in the point's directory	What it is
config.json, fsi-provenance.json fsi_nodes.csv	the effective configuration, and where every number came from the structural nodes, inside the component, in its own frame; one identical file imported once per blade
fsi_node_map.json fsi_family_map.json	the node ordering map (node_map_file), from the same generator the emitted section-load order: which rows of the loads export belong to which blade
fsi_post.txt	the native loads-export callback: update the sections, compute their loads in Newtons, export them
fsi_callback.py	the Python callback: one pyfs-fsi step in the point's directory

The callback runs under the interpreter that prepared the run (its pythonw.exe sibling on Windows): the package, with its [fsi] extra, must be installed for that interpreter on the machine that executes the point.

Source: pyflightstream [1]; docs/fsi-workspace.md, "Automatic workspace coupling"; cases/fsi_workspace.py, wire_workspace_fsi; fsi/beam.py (PyNite from the [fsi] extra).

The native commands, in the order they are emitted

Command	What the row gives it
DELETE_AEROELASTIC_STRUCTURAL_NODES	nothing: a clean start
AEROELASTIC_RBF_TYPE	MULTI_QUADRATIC, for a beam line of nodes
ASSIGN_AEROELASTIC_SURFACES	the coupled boundaries, by their identifiers in the mesh
ASSIGN_AEROELASTIC_COORDINATE_SYSTEMS	each blade's own frame, or the wing's
IMPORT_AEROELASTIC_STRUCTURAL_NODES	per blade, in blade order: its frame, DISABLE, fsi_nodes.csv
SET_AEROELASTIC_WORKING_DIRECTORY	., the point's directory
SET_AEROELASTIC_POST_PROCESSING_SCRIPT	fsi_post.txt
SET_AEROELASTIC_STRUCTURAL_EXECUTION_COMMAND	the interpreter and fsi_callback.py
SET_AEROELASTIC_ITERATIONS	1, one call per coupling step
EXECUTE_AEROELASTIC_ANALYSIS	a steady or qsteady_rotor row: the steady coupled solve
SET_AEROELASTIC_COUPLING_IN_UNSTEADY	an unsteady row: ENABLE, before START_SOLVER

fsi_callback.py, whole:

```
from pyflightstream.fsi.cli import main
raise SystemExit(main(['step', '--dir', '.']))
```

Documented, not verified

The command database records every command of the family as documented, none as verified: its 26.124 rows are carried from the 26.123 manual. The coupled runs of this version are measurements, not promotions.

Source: the requirements of this version; cases/fsi_workspace.py, wire_workspace_fsi; commands/aeroelastic_coupling.yaml.

The pproc sections a coupled row needs

A one-blade rotor whose family is Blade1:

```
[sections]
count = 50
include_symmetry = false

[[sections.distributions]]
families = ["Blade1"]
frame = "LOCAL_AXIS"
planes = ["XY"]
```

No separate [settings.aeroelastic] table is needed.

- Exactly one XY distribution per blade, emitted in blade order, each in that blade's own frame (span along Z); a wing takes one along its span.
- Several blades: select the rotor's blades so LOCAL_AXIS expands them in the declared blade order. The structural blade_count counts every explicit blade.
- Duplicate cuts, aggregated families, other planes or additional distributions are refused, never attributed to the first matching blade.

Takeaway *The loads export is one flat table in creation order; the family map is what tells the driver which rows are which blade.*

Source: pyflightstream [1]; docs/fsi-workspace.md, "Automatic workspace coupling"; FSI tutorial, "loads".

Part 5

What is refused, and why

Refused at plan: the shape of the run

The automatic route is deliberately bounded: each of these refuses the row when it is planned, before it can run.

The row	Why it is refused
turns a rotor (<code>unsteady_rotor</code>)	the plan says the feature is still in debug: a turning blade is deformed in its un-turned position [4]
is a <code>qsteady_rotor</code> wheel	the coupled quasi-steady case is the sector
continues a stopped run	the structural state and the section order of a continuation are not proved
opens a saved <code>.fsm</code>	the section order a saved simulation inherits has no proof: import the mesh, with its boundary sidecar
places nodes in a unit other than metres	other native node units have not been measured
carries a raw command or a custom flag	the section and frame order can no longer be proved
puts a structural node outside the component	the interpolation cannot place a surface around a node that is not within it

Source: the requirements of this version; `cases/fsi_workspace.py`, `validate_workspace_fsi`; `cases/workflows.py`.

Refused at plan: blades, frames and the speed

- **The count.** The structural `blade_count` must equal the rotor's blades, every one explicit.
- **The speed.** $\Omega = |\text{rpm}| \pi / 30$ must equal the row's rotor speed; on a `qsteady_rotor` row that is the speed of its free-stream rotation. A structure that does not turn has $\Omega = 0$ and carries its own weight instead.
- **The clock.** On an `unsteady` row the structural time increment must equal the solver's time step.
- **The mapping.** Blade, frame and boundary are one-to-one; there is exactly one section distribution per blade, in blade import order, each in its own frame with XY cuts, span along Z.

Geometry and structural pitch must describe the same blade: the adapter infers no basis and re-pitches no mesh.

Source: the requirements of this version; `cases/fsi_workspace.py`, `wire_workspace_fsi`, `validate_workspace_fsi`.

Refused while reading the input

The input	The message says
a factor key the package does not know	"unknown FSI matrix keys [...]; use FSI_GEOMETRIC_PITCH_DEG_FACTOR, ..."
factors on a row with no FSI	"matrix calibration states no FSI input; add FSI: f001 in VAR_NAMES_VALUES."
a code that is not f + name	"FSI='...'; name an f-prefixed code without an extension, such as f001."
a zero or negative factor	"... use a finite positive factor."
a stiffness scale other than 1 in [config]	"... config.stiffness_scale_factor must be 1.0 in workspace inputs; calibrate bending_stiffness_n_m2 and torsion_stiffness_n_m2 explicitly ..."
material factors in supplied mode	"supplied mode cannot apply material factors [...] to already supplied distributions; calibrate the distribution itself."
a blade table in calculated mode	"calculated mode derives config.blade; remove the supplied blade table."
a material the database lacks	"no material '...' in the materials database (version 1); it holds: al-7075-t6, ti-6al-4v-grade5-annealed. ..."

Source: `pyflightstream [1]; workspace/fsi_setup.py, resolve_row_fsi, resolve_fsi_setup; fsi/materials.py, material.`

Part 6

Worked examples

Example 1: calculated and supplied inputs agree

examples/workspace_fsi_calibration.py,
offline, no solver:

1. Writes inputs/fsi/f_calculated.toml from the package template (the titanium strip), with the file factor 1.05 on EI.
2. Resolves it and reads the *unscaled* base distributions.
3. Writes f_supplied.toml with those base distributions and the same file factor 1.05. Reusing the effective values here would apply the calibration a second time.
4. Resolves both with
FSI_BENDING_STIFFNESS_N_M2_FACTOR: 1.15.
5. Asserts every distribution equal across the two modes, effective EI equal to base EI times 1.15, and the origin matrix.

```
python workspace_fsi_calibration.py /absolute/path/to/new-  
workspace
```

Source: pyflightstream [1]; examples/workspace_fsi_calibration.py, run at the tag; workspace/fsi_setup.py, FSI_TEMPLATE.

Printed	Value
base EI, each station	24.277 N m ²
effective EI, each station	27.919 N m ²
file factor, matrix factor	1.05, 1.15
factor_origin	matrix
mode_equivalence	verified
solver_executed	false

Check it by hand

$$I = bh^3/12 = 0.04 \times 0.004^3/12 = 2.133 \times 10^{-10} \text{ m}^4; EI = 113.8 \text{ GPa} \times I = 24.28 \text{ N m}^2; \times 1.15 = 27.92, \text{ not } \times 1.05 \times 1.15.$$

Choose a new destination: the example refuses to overwrite its two inputs.

Example 2: one section, and its torsion constant

examples/fsi_solid_blade_properties.py builds a synthetic blade: seven stations from 0.15 to 0.75 m, chord 80 to 40 mm, pitch 35 to 11 deg, NACA 2415 at the root thinning to NACA 2409 at the tip, written to a section file and read back, the file's sha256 into the provenance.

Root section, NACA 2415

area	654.144 mm ²
centroid from the pitch axis	-13.422 mm chordwise
I flap, I chord	5512.2, 227784.7 mm ⁴
principal angle	-0.100 deg

Grid, cells across	J [mm ⁴]
16	20926.02
32	20956.81
64, the default	20962.52
thin-section estimate	21721.14

- Doubling the cells divides the error by about four: at 64 cells about a third of the change from 32 to 64 is left.
- The thin-section formula is 3.6 % high here: a cross-check, never the value.

Source: pyflightstream [1]; examples/fsi_solid_blade_properties.py, run at the tag.

Example 2: the whole blade, in titanium

r [m]	μ [kg/m]	EI [N m ²]	GJ [N m ²]	I_1 [kg m]	I_2 [kg m]	e_c [mm]	thin/ J
0.15	2.8979	627.29	922.35	1.009e-03	2.442e-05	-13.42	1.036
0.25	2.2727	361.01	531.76	6.650e-04	1.405e-05	-12.30	1.032
0.35	1.7441	198.04	291.97	4.217e-04	7.708e-06	-11.19	1.028
0.45	1.3040	102.60	151.24	2.554e-04	3.993e-06	-10.07	1.024
0.55	0.9445	49.58	72.98	1.462e-04	1.930e-06	-8.95	1.020
0.65	0.6574	21.98	32.24	7.789e-05	8.553e-07	-7.83	1.017
0.75	0.4347	8.73	12.72	3.784e-05	3.395e-07	-6.71	1.014

- Ti-6Al-4V grade 5, annealed: ρ 4430 kg/m³, E 113.8 GPa, G 44.0 GPa, ν 0.342, database version 1.
- The configuration round-trips through `config.json` unchanged, with its provenance and its `config_sha256`.
- e_c : the centroid ahead of or behind the pitch axis, which the elastic axis is assumed to follow.

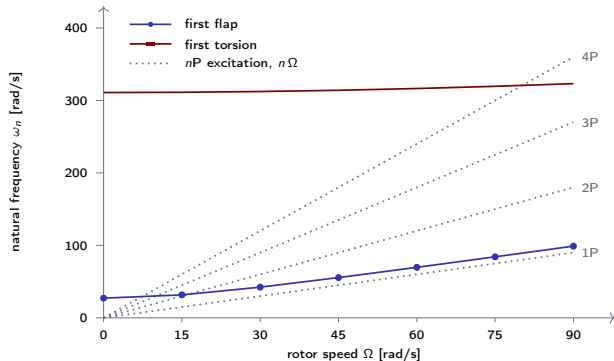
At rest, from the beam

first flap: 30.6 Hz

first torsion: 601.9 Hz

Source: `pyflightstream` [1]; `examples/fsi_solid_blade_properties.py`, sections 3 and 4, run at the tag.

Example 3: the Campbell diagram of a rotating blade



A uniform synthetic blade, 1 m, 15 stations, swept from 0 to 90 rad/s. At every speed: static solve under its own centrifugal loads, then the modes with the tension's geometric stiffness.

Southwell fit	flap	torsion
ω_0 [rad/s]	27.96	310.90
S	1.118	0.961
r^2	0.99986	1.00000

$\omega_n^2 = \omega_0^2 + S \Omega^2$. Flap S sits in the expected 1.1 to 1.3; torsion near $(I_1 - I_2)/(I_1 + I_2) = 0.961$. A crossing of a track and a ray flags a potential resonance [10, 11].

Source: `pyflightstream` [1]; `examples/fsi_campbell_diagram.py`, run at the tag; FSI tutorial, "centrifugal".

Part 7

Worked example: the XPROP rotor in titanium

Example 4: the XPROP rotor as solid titanium

- The XPROP rotor [2], six blades, $J = 1.70$, 473.2 rev/min, Mach 0.144; unsteady, 10° per step, four revolutions, far field 5 layers; build 26.124 [3].
- The blades are **solid Ti-6Al-4V** from the package database: ρ 4430 kg/m³, E 113.8 GPa, G 44.0 GPa (tabulated), ν 0.342; sections calculated from the blade contours (inputs/fsi/f001.toml, calculated mode).
- The **stiffness gain** K scales EI and GJ together through the matrix: FSI_BENDING_STIFFNESS_N_M2_FACTOR and FSI_TORSION_STIFFNESS_N_M2_FACTOR = 0.5, 1.0, 2.0; mass and inertia unchanged. Provenance records both with origin `matrix`.
- The comparison: the same rotor with **rigid blades**.

Source: pyflightstream campaign, 2026 [3]; the campaign's record.

The matrix, in one line each

rigid: no FSI column

titanium: FSI: f001, both factors = K

One structural file, three rows: the gain is a matrix variable, never a new input file.

Example 4: solved by route B, because the coupled route failed here

What the earlier command sequence did here

The three coupled rows stopped at the first coupling call: the solver reported the RBF weights singular in the LU decomposition and exited. The node file imported once per blade frame leaves coincident nodes on 26.124 for a rotor of more than one blade. It was a defect of that command sequence; none of those rows is a result.

- **Route B** instead (Part 1): the package's own chain (`to_elastic_axis`, `solve_rotating_static`) on the native sectional loads gives flap $w(r)$ and twist $\theta(r)$.
- Every mesh vertex moves about the elastic axis by that w and θ ; the solver reruns the deformed rotor as the baseline.
- Stop: tip flap changes by less than 1 % and tip twist by less than 0.005° between two structural solutions.
- Control: a zero deformation through the same pipeline reproduces the rigid rotor to 2.4×10^{-5} in thrust.

Source: pyflightstream campaign, 2026 [3]; `fsi_route_diagnostic.json`.

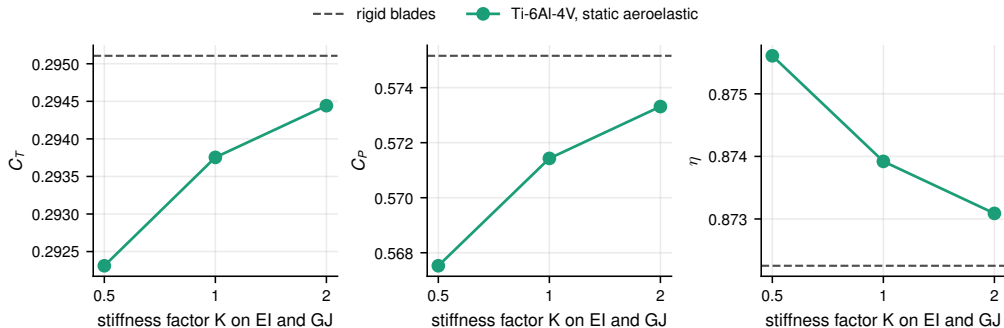
Example 4: what the earlier sequence did, and what changed

configuration on 26.124	earlier planner	loop runs	deforms	rotation
six blades, product route	accepted	no: RBF singular		
one blade, no symmetry, product route	accepted	yes, 81 calls	no	yes
one blade, periodic copies (a sector)	refused			
one blade, 5 cm prescribed, rotating	(probe)	yes	no	yes
one blade, 5 cm prescribed, no motion	(probe)	yes	no	
fixed wing, steady	refused	callback never called	no	
fixed wing, unsteady, 0.2 m prescribed	refused	yes, every step	no	
fixed wing, unsteady, coupled beam	refused	yes, 30 of 30	no	

- With no motion at all, a 0.2 m bend and a 5° twist leave the loads identical to every printed digit: the displacement never reaches the aerodynamic surface, and motion is not the cause.
- Scope: the earlier command sequence, on 26.124. This version builds the nodes inside the blade and assigns the surface by its boundary identifier; with them a blade held still in a turning free stream maps and deforms (the `qsteady_rotor` sector), while a turning blade is still deformed in its un-turned position, so `unsteady_rotor` with FSI is refused [4].

Source: `pyflightstream` campaign, 2026 [3]; `fsi_wing_probe.json`; `fsi_route_diagnostic.json`.

Example 4: performance against the stiffness gain

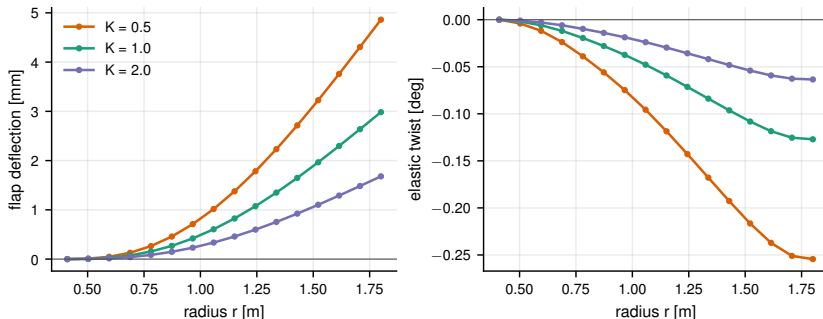


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case	T [N]	C_T	C_P	η	ΔC_T [%]	$\Delta \eta$ [pts]
rigid blades	2078.0	0.2951	0.5752	0.8723		
Ti, $K = 2.0$	2073.3	0.2944	0.5733	0.8731	-0.23	+0.08
Ti, $K = 1.0$	2068.4	0.2938	0.5714	0.8739	-0.46	+0.17
Ti, $K = 0.5$	2058.3	0.2923	0.5675	0.8756	-0.95	+0.34

Source: pyflightstream campaign, 2026 [3]; fsi_xprop_comparison.csv.

Example 4: how the blade deforms

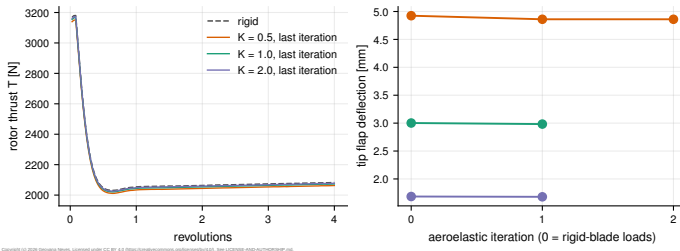


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- The elastic twist is **nose down** and scales as $1/K$: it unloads the outer blade, so thrust and power fall and efficiency rises.
- The flap scales **less** than $1/K$ (tip 1.68, 2.98, 4.86 mm): centrifugal tension stiffens the blade in flap, as the rotating beam says.

Source: pyflightstream campaign, 2026 [3]; `fsi_xprop_spanwise_deflection.pdf`.

Example 4: convergence, and what the example establishes



- One or two aeroelastic iterations suffice; the one-way answer (rigid-blade loads) is already within 1.3 % of the coupled flap.
- The change against rigid is stable to about 0.01 points over revolutions 2 to 4.

What it does not establish

Native two-way coupling; dynamics (the iteration is static); absolute C_T better than 0.45 % (thrust still rises that much per revolution). No spinner or nacelle: rigid thrust 3 % below the same case run with them.

Source: pyflightstream campaign, 2026 [3]; [fsi_xprop_convergence.pdf](#).

Part 8

Outputs, checklist and limits

Outputs, all in the point's working directory

File, written at every call	What it holds
FS_SurfaceSection_Loads.txt	by the solver: the section loads, computed in Newtons, line densities along the span; the reader asserts units and completeness
FSIDisp.txt	one translation per node, in metres, blade frame, import order
state.json	the structural state, written atomically, so a killed call resumes
fsi_convergence_log.csv	the validity boundary in its header, config_sha256 on every row
fsi_archive/call_NNNN/	that call's loads and displacements, for offline replay
pyfs_fsi_calls.log, _error.log	a line per call; on failure the traceback, and a non-zero exit that stops the coupling rather than let it continue rigid

- `pyfs-fsi step --dir /absolute/path` runs one driver call when current loads and maps already exist; stale solver iterations are refused.
- Changing a calibration requires a new resolved point and regenerated nodes.

Source: `pyflightstream` [1]; `docs/fsi-workspace.md`, "Automatic workspace coupling"; `fsi/cli.py`; `fsi/driver.py`; FSI tutorial, "loads", "driver and state", "cli".

Before planning an FSI row

1. **The input:** `inputs/fsi/f<id>.toml` in ONE mode, SI units; calculated mode names a database material or a fully sourced one, and closed simple contours in metres about the pitch axis.
2. **The row:** `WORKFLOW` steady, unsteady, or `qsteady_rotor` with periodic symmetry (the sector); FSI: `f<id>` and any `FSI_*_FACTOR` in `VAR_NAMES_VALUES`; no RAW, no custom flags.
3. **The geometry:** a fresh OBJ or STL import with its boundary sidecar, in METER; never a saved `.fsm`.
4. **The rotor:** one isolated, axisymmetric rotor; shaft X, blade datum Z; `blade_count` equal to its blades; `omega_rad_per_s` equal to the row's free-stream rotation. A wing: its span, and $\Omega = 0$.
5. **The pproc:** one XY distribution per blade, `LOCAL_AXIS`, in blade order, `include_symmetry = false`.
6. **Then** `pyfs-matrix` plan, where the refusals of this deck fire; after the run, `fsi-provenance.json` in the point's directory gives each factor's value and origin.

Source: `pyflightstream` [1]; `docs/fsi-workspace.md`; `cases/fsi_workspace.py`; `workspace/fsi_setup.py`.

Limits: what the coupling does not establish

- **Wiring is not validation:** the staging and callback checks do not establish native two-way rotor deformation, convergence or aerodynamic validity.
- **The native evidence so far** [4]: on 26.124 a blade held still in a turning free stream, its nodes inside the blade and the multiquadric kernel, mapped every node and deformed; its thrust stayed within 0.35 % of the rigid turning blade. A turning blade is deformed in its un-turned position, which is why `unsteady_rotor` with FSI is refused.
- **Open:** saved-simulation and continuation use, and native coupled validation; staging or command acceptance implies neither.
- **The zero-speed example inputs** are for offline checks, not runnable coupled rotor rows.

The package's own capability table

FSI beam and modal analysis: experimental. The coupled driver: experimental, replayed offline on archived fixtures, never run against a live solver in CI. Rotary two-way coupling: not validated.

Source: `pyflightstream` [1]; `docs/fsi-workspace.md`, "Automatic workspace coupling"; `README.md`, the capability status table; `docs/srs/architecture-srs.md`.

Limits: the structural model

- **An Euler beam**, one per blade, in flap and torsion (w, θ): only those two carry mass. This input adds no shell, spar, hollow-section, shear-centre or second structural model; hollow and spar sections are a future option, not built.
- **Solid, homogeneous sections** from ONE material, when calculated.
- **The elastic axis at the centroid**; the shear centre is not computed. A generated blade has a zero CG offset, so the bend-twist coupling that offset drives is absent.
- **No trapeze effect**: the elastic twist is overpredicted, which is conservative.
- **Quasi-static**: the once-per-revolution content is trustworthy only well below resonance, $n\Omega/\omega_n \lesssim 0.3$.
- **Sources**: the formulas are transcribed from the coupling plan, not re-checked against the primary papers; the materials are typical values, not design allowables.

Source: pyflightstream [1]; docs/fsi-workspace.md; FSI tutorial, "The loop in one paragraph", "sections and materials", "beam", "centrifugal"; fsi/materials.py.

The physics behind the coupling, and where to read it

- **The beam:** Euler bending and Saint-Venant torsion of a slender member, and the section constants of a solid contour [6, 7].
- **The rotating blade:** centrifugal tension and stiffening, the propeller moment toward flat pitch, the in-plane softening, and the natural frequencies that rise with the speed [8, 9, 10].
- **The Campbell diagram:** the blade's frequencies against the rotor speed, beside the lines of the rotor's harmonics [11].
- **Quasi-static coupling:** the unsteady aerodynamics of an oscillating section, and the reduced frequency below which a steady answer holds [12, 13].
- **The interpolation:** radial basis functions from a set of structural nodes to an aerodynamic surface, and the multiquadric kernel [14, 15, 16].

References

- [1] Geovana Neves, *pyflightstream documentation*, page “FSI in a workspace”.
<https://nevesgeovana.github.io/pyflightstream/fsi-workspace/>; and the FSI tutorial,
[src/pyflightstream/fsi/README.md](https://github.com/nevesgeovana/pyflightstream/blob/main/src/pyflightstream/fsi/README.md) of the repository.
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